Computational Intelligence, Fuzzy systems, and Machine Learning: academic vs industrial learning

Roman Neruda[1] and Juan Carlos Figueroa-García[2]
[1]*Institute of Computer Science, Academy of Sciences of the Czech Republic, Prague - Czech Republic*
[2]*Universidad Distrital Francisco José de Caldas, Bogotá - Colombia*

***Abstract*—Computational intelligence, fuzzy systems, and machine learning have become very popular in different academies around the globe, including mathematics, statistical, computer science, philosophy departments, etc. While a large body of academic research has been done in the last 50 years, industrial and real world applications are incipient when compared to the amount of people involved in academic research. Thus, there are many questions and complaints to be answered in order to apply all we (scientists) have done. To do so, we provide a short-comprehensive diagnosis of the current situation and provide some short term strategies to improve the link between industry applications of academic research.**

***Index Terms*—AI, Machine Learning, Fuzzy systems, Industrial applications**

I. Introduction and Motivation

COMPUTATIONAL intelligence *(CI)* appeared as a fuzzy word in the late 70's when computing capabilities came from very basic operations to most sophisticated routines which opened a door to learn how to control a computing machine. Even in the early years, there was a clear difference between hard and soft computing in the sense that computer scientists had to work in two directions: how to build better Turing machines and how to do better programs.

Soft computing started to make sense since Lofti A. Zadeh published his seminal paper [1], and later in [2] when he defined the role of neural networks and fuzzy logic into soft computing. At the same time, Bezdek & Pal [3] have defined CI as a wider field that involves many other intelligent techniques (hard and soft) which finally led into IEEE-CIS society. In different conferences such as NAFIPS, SMC, IFSA, WCCI, FUZZ-IEEE, ICPR, ISNN etc, there have been discussions about the role of CI, fuzzy sets and machine learning in industrial applications, and some of the main concerns of academic community regard to a lack of connection among academics, practitioners, industry, and how to efficiently face up future problems.

The relationship between academic research and industrial research in CI topics is somehow clear, and one of the most important concerns goes to how to interrelate a huge amount of academic developments to practical applications (which include industrial, public sector and common life)? This way, we addressed the problem of interrelate base education to scientific studies in order to finally obtain successful applications of CI, fuzzy sets, and machine learning.

The paper is organized into 6 sections; Section 2, 3 and 4 presents a discussion about the current status of CI, fuzzy sets and machine learning techniques in academic and industrial fields; Section 5 shows some strengths and weaknesses of every topic; In section 6, some recommendations for the future are given, and finally Section 7 presents the concluding remarks of the study.

II. CI current status

Most of the people recognizes the term *Artificial Intelligence (AI)* and immediately think on the 80's movie AI; other people think on Japanese robots playing soccer or dancing some music; other people remember some Hollywood movies where wars are ran in the future using AI, autonomous warcrafts and so on. The question is: do we have the technology to create a machine that provides intelligent and autonomous solutions to a problem? Well, the answer is: no.

A more realistic term to represent the current status of what humans have developed in the last 50 years regarding non human intelligence is the term *Computational Intelligence (CI)*, because we can create intelligent computer programs so far, not autonomous machines (see Zadeh [2] and Bezdek & Pal [3]). Are we far from general AI? Probably no, since military applications of CI have reached an interesting level, of course its current state is not opened to the public. Just have a look of the new F-35, or the last Israeli missile technology in order to have a good example of applications of CI to pseudo-autonomous machines.

Now, what about industrial applications of CI?. There are many applications of some selected topics such as neural networks to predict stock prices and exchange rates, agent based simulation to market analysis, pattern recognition on internet applications, etc. Most of those applications have been developed by scientists to world size organizations such as Google, Microsoft, Xbox, Playstation, Huawei, Samsung, KIA motors, Hyundai, Apple, NYC Stock Market Inc., and some customer analysis companies who offer specialized services

Roman Neruda is Assistant Professor of the Inst. of Computer Science of the Academy of Sciences of the Czech Republic, Prague - Czech Republic, e-mail: roman@cs.cas.cz

Juan Carlos Figueroa is Assistant Professor of the Engineering Dept. of the Universidad Francisco José de Caldas, Bogotá - Colombia, e-mail: jcfigueroag@udistrital.edu.co

978-1-5090-0228-3/15/$31.00©2015 IEEE

as third party of big companies such as MacDonalds, KFC, Haggen Daas, Coca-Cola, etc.

The current status of CI from an academical perspective is not what we (scientists) have expected. There are many students who taken CI related courses, many outstanding students who have done something new and interesting during their bachelors, many brilliant scientists, businessmen, philosophers, managers, engineers, etc. who have pushed science to a new level in different ways, but not many industrial applications in low-medium size companies have been reported. So, what is going on low-medium companies? Are they just using basic tools to solve their problems? Why do not they want/wish to apply CI to their problems?

What we meant in last paragraph is: why many CI academic developments are reported, but only a small amount of applications to low-medium size companies have been reported?. Is CI only attractive for scientists? How we (scientists) can help to popularize the application of CI to low-medium size companies' problems?. We think that this problem is out of CI and industrial scope, it is a *communication problem* since most of good scientists have totally different characteristics from managers, decision makers, analysts, etc.

One of the most important skills in human being interaction is *communication*, so we (scientists) can do to help popularizing our results and do a wider application of our developments is to find the way to improve our and our students communication skills, but how to do it?. It is a hard question because it involves many human being features such as personality, humor, gifts, mentality, prejudices, cultural roots, religion, etc. What it is clear is that scientists need to improve their communication skills in order to transfer knowledge to a wider audience. We (scientists) need to understand our customers needs, and efficiently communicate our ideas for solving their problems.

On the other hand, there is another problem that CI scientists face: the problem of finding the simple way to solve a problem. Sometimes, the best way to solve a big problem is by finding a simple way to do it, and sometimes small problems are intended to be solved by using sophisticated methods when it is unnecessary. We (scientists) need to acquire the ability to recognize any problem size and propose a well sized solution to. But, how to do it? Well, what we recommend is to share experiences and learn from them, as neural networks do. Seminars, forums with industry people, visits to low-medium sized companies (public and private sectors) are a good starting point to get the abilities to recognize and understand what they need.

III. Machine learning current status

The ultimate goal of machine learning is to build a computer system that can extract relevant information from data, and act on it without being explicitly programmed by humans for a particular task. The autonomous way of data-driven operation – be it a medical diagnosis software, or a robot controller – is achieved by general model building capabilities of approaches based on statistics, search algorithms, and computational models from support vector machines [4] to neural networks [5]. Traditionally, the tasks performed by ML are categorized as supervised learning (classification or regression), unsupervised learning (clustering or density estimation), and reinforcement learning (where a system interacts with the environment, and tried to maximize a reward based on the actions performed in the environment).

Traditional AI approaches to ML include methods like inductive logic programming, expert systems, decision trees, or association rules learning [6]. They are commonly based on formal calculus, logical knowledge representation, and various reasoning techniques. This family of algorithms has recently undergone an extensive grow with the relatively new field of multi-agent sytems [7] where various planning and action selection mechanisms co-exist. The formal knowledge representation is also crucial for the ontological systems that try to model semantic relationships of concepts and roles. The emphasis on semantic structure is currently very visible in the areas of multi-agent systems [8] or web-based information, especially the so-called semantic web [9]. Such approaches are quite popular, because their knowledge representation is relatively understandable to humans, the reasoning procedures have sound mathematical background, and provide results that are easy to interpret.

CI methods applied to ML have quite several opposite properties [10]. The neural networks or support vector machines [11] are very good regression and classification tools. They have been used in successful applications on problems like optical character recognition, or as data-driven controllers. The disadvantage is the distributed and subsymbolic knowledge representation, which prevents us to understand the meaning of the model most of the time. Neural networks have become popular recently with the spread of deep learning [12] approach that is now able to work with networks of dozens layers and thousands of units. An interesting factor that allowed people to experiment with such big models, was the availability of GPU hardware which has demonstrated to be very suitable for this type of computations.

In recent years we have in fact experienced several success stories in the area of ML. The convolutional deep networks [13] – which are specialized to work on image data– have achieved very good performance in the task of image and video annotations. Different deep networks are able to learn to play simple computer games or analyze natural language [14]. Another large and complicated machine learning procedure has been applied to movie recommendation task in the so-called Netflix prize [15]. The winning team has worked on a solution for two years, and the resulting model has several hundreds of carefully chosen models into various combinations. The challenge of today is to provide this level of modelling complexity by means of adaptive and learning algorithms only, thus replacing human experience with machine intelligence and adaptation.

IV. Fuzzy systems current status

We are celebrating 50 years of publication of the Lofti A. Zadeh's seminal paper [16] that opened a door to think in *Soft computing* rather than computing in a soft way as we did in the early years (when human machine interfaces started to be important in computer science). What Lofti A. Zadeh did was to think over machines in order to propose a way to represent natural language which involves uncertainty by nature, so many people started to think different after reading their papers (just for the record, Elie Sanchez [17], M.G. Thomason [18], H. J. Zimmermann [19] among others). Thus, a third player appeared in the game: soft computing as a new scientific field that offered new tasks and challenges to computer science.

Many years (in a fuzzy timeline) have been passed until new developments saw the light of publication. For instance, K. Atanassov [20] proposed his intuitionistic approach based on complementarity between two membership functions, Kreinovich et al defined a clear link between interval computations and fuzzy sets [21], but definitely one of the most important advances in fuzzy set theory so far is the definition of Type-2 fuzzy sets given by Jerry Mendel [22] who opened a new door in natural reasoning when multiple people use words and concepts to solve real world problems.

All those results and advancements have been achieved in an *academical environment*, but what is going on real world applications? There is no point if we try to compare flat numbers between the amount of fuzzy scientists and fuzzy industrial implementations of fuzzy techniques because real world applications would loss the game, but we can think on what is happening in fuzzy community nowadays. Why fuzzy techniques are not implemented as we expected? Why so many scientists show a real interest on fuzzy theory but most of their results does not end in industrial applications but in a technical report?

On the other hand, acceptation of fuzzy theory in different universities and schools have been very strong from the beginning since it was really well defined and supported on information theory and uncertainty theory. Zadeh's ideas never went to try to replace probability theory but complement their advancements and scope, which had a real scientific spirit so many people started to see how fuzzy sets (and possibilistic uncertainty) complement probabilistic uncertainty in a good way. But why probabilistic approaches are widely applied while fuzzy techniques (even when fuzzy uncertainty generalizes crisp and probabilistic set theory) are not?

Form an academic point of view, fuzzy sets and systems are easy to understand and they can be widely taught in schools and bachelor degrees without any resistance from students, at the same time than probabilistic concepts. What is the idea then? Well, as human beings can learn as much as we teach them in early ages, then the idea is to popularize uncertainty analysis on early learning stages in order to familiarize people to uncertainty. Evidently the idea is not to replace probabilistic thinking but induct multiple reasoning thinking into the people's mind, in other words, we need to start a paradigm shift in teaching: a multiple paths teaching structure.

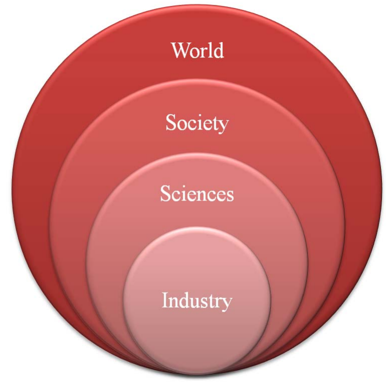

Fig. 1. Teaching in different scenarios

Figure 1 shows how teaching influences industry, society, and sciences. Different teaching schemes are applied in every stage, so articulation among stages could lead to a better understanding of CI, fuzzy systems and machine learning. This also leads to better implementations by reducing risk and increasing people's knowledge on exact sciences, computer science, natural reasoning, approximate reasoning, human-like knowledge, etc. While education for the society should be oriented to understand the world, live in society, and how to coexist in the environment which implies exact, natural, social, environmental sciences; academic and industrial teaching should lead to research about how to build a sustainable society.

In general, not only industrial applications need experts on CI, fuzzy systems and machine learning; education for living in a technologically-oriented world needs it as well. This way, there is a need for improving education in a modern world in order to understand how technology works and impacts our environment in short and long terms.

V. Strengths and weaknesses

The main problem of applying machine learning and computational intelligence to solve practical tasks is the heterogeneous nature of their approaches. The methods coming from different fields of mathematics, statistics and computer science can all be used to solve a vast spectrum of problems, and the model selection is often a matter of engineer's knowledge and experience. Moreover, a deep, and quite general theoretical result – the No free lunch (NFT) theorem [23] – states that the performance of all machine learning models will be the same when averaged over all the possible problems. This should motivate us to create a particular optimized data-driven model for each task we encounter. Among the current and promising approaches how to tackle the complexity of real-world applications of AI in general are deep learning, hybrid solutions, and meta-learning.

Deep learning and deep networks [12] is a hot topic in AI with several breakthrough applications in recent years. While many ideas in deep networks can be seen just as an evolutionary step in a half a century of artificial neural networks

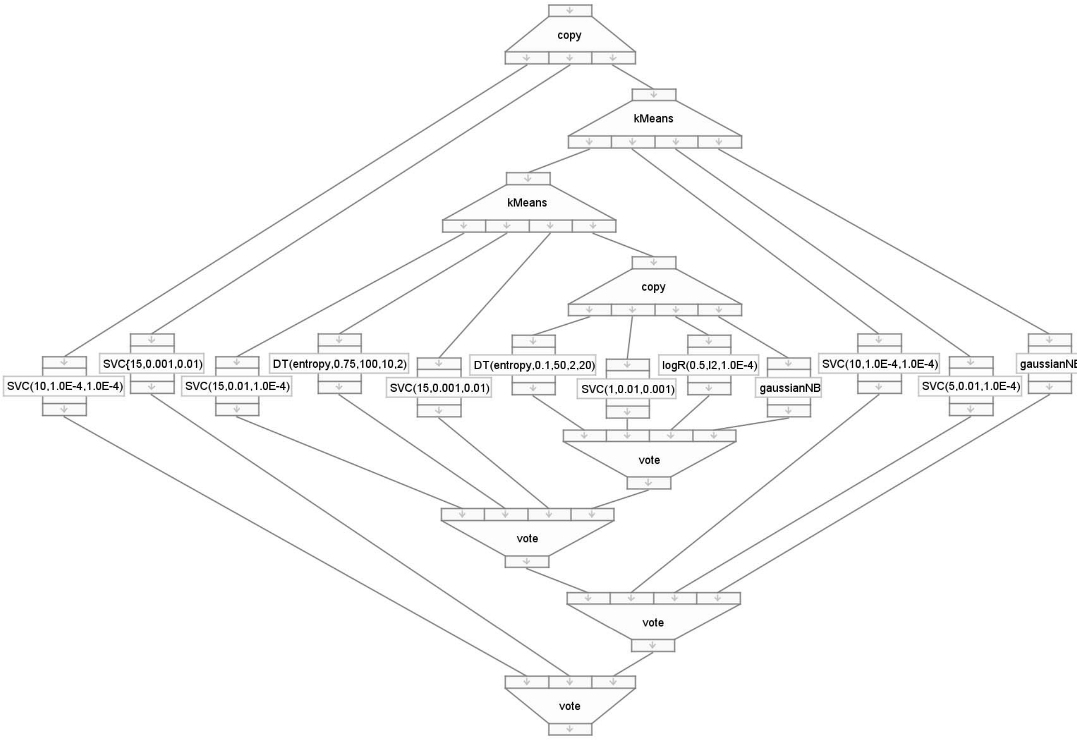

Fig. 2. A scheme of the best autonomously found workflow for the wilt benchmark data set.

research, the combination of suitable hardware platforms (the graphical processing units, or GPUs) and specialized network architectures represent the necessary combination towards successful engineering solutions. The specialized convolutional nets [13] have performed surprisingly well in the areas of image classification and pattern recognition. These large autonomous models are now providing human-competitive results in image and video analysis where the large amount of engineering experience and empirical skills is surpassed by clever algorithms and vast computer processing time. Besides the enormous computational time, deep networks also suffer from the problem of various possibilities of units and layers that need to be composed in a reasonable manner in order to solve a given problem efficiently. Nevertheless, the field is quite new and these problems are tackled currently, and we expect more results in this area, maybe by means of hybrid approaches that are now efficiently applied in the areas of more traditional machine learning models.

Hybrid approaches [24] and combinations of methods have always been an interesting area of research, the ideas were in the very core of soft-computing approach and are now providing common ground for computational intelligence and fuzzy combinations. We have already mentioned the Netflix prize winning solution containing hundreds of individual models from decision trees to linear and Bayesian approaches, combined in a non-trivial way by means of ensembles of experts. This is a typical example of hybrid solution. Another might be to use evolutionary algorithms (EA) [25] to set parameters of a model by means of a robust search procedure. EAs can be successfully combined with neural, fuzzy or reinforcement approaches to add learning and adaptation to other well established modeling techniques.

The problem of many of the above mentioned methods is that they typically encompass a non-linear optimization procedures in high dimensional spaces. Thus, the algorithms are quite sensitive for setting the initial condition and to particular values of different parameters of the methods, usually referred to as hyper-parameters [26]. These may represent the size of the model, many convergence or regularization related parameters, but even things like a choice of metrics, choice of kernel function, and similar. Again, while some heuristic and empirical approaches can be applied, the setting is data-dependent and should be treated individually for every task to be solved. It is interesting to apply different search techniques from hill-climbing to EAs in order to find a good hyper-parameter setting, maybe taking into account the previous experience with model performances.

This is one of the tasks of the approach called Meta-learning [27] that tries to improve learning process by applying machine learning procedures to the data about the models, their settings and performance. Besides hyper-parameter settings, the main goal of meta-learning is the model recommendation. We have already mentioned the NFT theorem which stresses the necessity to consider data-dependent modeling. Meta-learning approach to model recommendation considers the similarity of data sets representing the tasks to be solved, and a history of performance of sufficiently many models on sufficiently many data sets. When the new data set is presented,

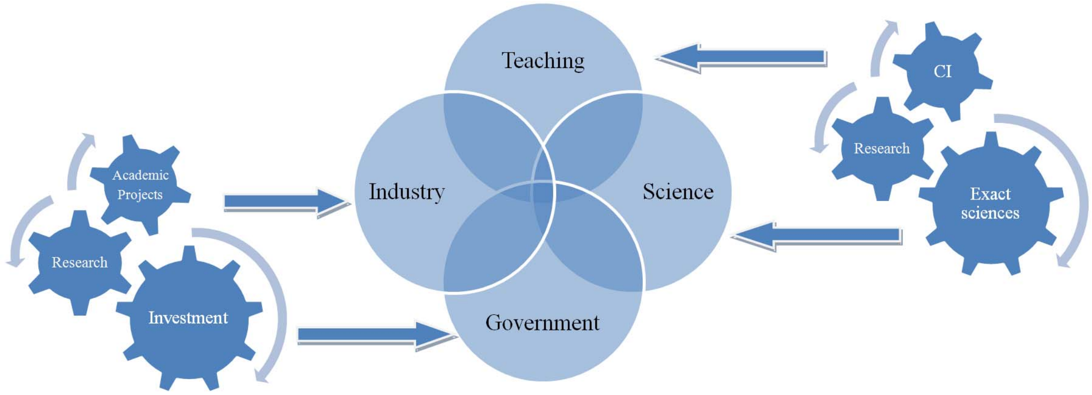

Fig. 3. Roles and relationships among actors

the recommender first identifies the most similar data sets from the historical data (this is called zooming), and then explores the performances of models on these data (this is called ranking). Finally, it recommends the model with the best performance on the zoomed neighborhood.

The next natural step that takes the above mentioned approaches together is to generate the whole workflows, or combination of methods with preprocessing algorithms, and proper parameter settings, to provide a tailored hybrid solution for a given task. A solution that should be comparable to the Netflix prize example with dozens or hundreds of cleverly combined components. As an example, let us show one such an approach that uses a special type of EAs, the so-called genetic programming technique [28] that searches the space of workflows and by means of local optimizations tries to find optimal solution scheme and its settings [29]. On Figure 2 we show a resulting workflow containing several machine learning and preprocessing models connected into several ensembles. This workflow has been evolved for the wilt data set from the UCI machine learning repository, and it outperformed several hand-made solutions. The amount of processing time to achieve such a solution is rather big, the whole algorithm requires the order of magnitude of thousand hours of computing time on one processor core.

Figure 3 shows how different actors such as government, industry, academy (science), and teaching (education) should interact together to obtain better results. Investment, research and academic projects should be important activities of actors like government and industry, but in most of cases they are absent from many organizations. On the other hand CI, research, and exact sciences should be an important part of science and teaching, but we need to improve and increase educational topics and intensities.

In a modern society, it is necessary to educate people understand technology and how to use it, not only to use it. If we start to teach people computer science, CI, math, chemistry, environment sciences, etc, then industrial applications for a better life will come as a natural step.

VI. Future work

Modern life technology is mostly user-oriented, but not many users understand how technology works. A possible (probable) explanation of this behavior is that education in basic sciences such as mathematics, philosophy, chemistry, etc is still limited, so most of people involved in industry have no a good knowledge about science instead of technology.

Technology usage is important, and understand how technology works is even more important. Is AI possible (in the future)? Yes, but there is a need to make CI, Machine Learning and soft computing wider and easy to understand to most of future users. This is only possible via popularization of CI, machine learning and soft computing techniques and concepts, not only in specialized rooms but in common life and industrial applications.

Industry faces some important challenges: how to invest in academic projects? how to obtain profits from their developments? how to get into a tech-society by offering low-cost high level solutions?. And the most important, how to become together as one with scientists?

On the other hand, (we) scientists have the challenge of doing a better approaching to industrial and real life problems. This means that we need to open the window of our laboratories (at least one time) to see real world and connect to practical unsolved problems. Efficient communications with industry, public sector and common people can help to understand problems and teach science in a better way.

VII. Concluding Remarks

We have identified several challenges and pitfalls in the area of academic vs. industrial approaches to computational intelligence, fuzzy systems, and machine learning. We believe there is a still a lot of work to be done to transfer the exciting research topics to industry, and for the time being we have discussed and proposed several points.

The engineers should be aware of the fact that there is no panacea, or silver bullet that will solve any problem. It is really important to always choose the appropriate data-dependent model to solve the given task. Since the human experience is rather limited, and the methods are very diverse, clever recommending algorithms are really important.

The use of meta-learning to propose models based on previous experience is a state-of-the-art solution to this problem. We have demonstrated, that a model recommendation is not really enough, and in the future we will see more recommendations of the hybrid solutions and workflows encompassing the whole data modeling process from feature selection, preprocessing, to ensembles of models.

And last but not least, the search for right setting of hyperparameters is crucial in order to achieve good performance of the model, especially in the case of more complicated solutions that require a lot of computing time.

Acknowledgements

R. Neruda has been supported by the Czech Science Foundation project no. GA15–18108S.

J.C. Figueroa-García would like to thank to all members of LAMIC research group and IEEE Student Branch of the Universidad Distrital Francisco José de Caldas.

References

[1] L. A. Zadeh, "Fuzzy sets," *Information and Control*, vol. 8, no. 1, pp. 338–353, 1965.

[2] ——, "Fuzzy logic, neural networks, and soft computing," *Communications of the ACM*, vol. 37, no. 3, pp. 77–84, 1994.

[3] J. C. Bezdek and S. K. Pal, *Fuzzy models for pattern recognition*. USDOE Pittsburgh Energy Technology Center, 1994, ch. 1.

[4] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995. [Online]. Available: http://dx.doi.org/10.1023/A:1022627411411

[5] S. Haykin, *Neural Networks: A Comprehensive Foundation*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall PTR, 1998.

[6] R. Agrawal, T. Imieliński, and A. Swami, "Mining association rules between sets of items in large databases," in *Proceedings of the 1993 ACM SIGMOD International Conference on Management of Data*, ser. SIGMOD '93. New York, NY, USA: ACM, 1993, pp. 207–216. [Online]. Available: http://doi.acm.org/10.1145/170035.170072

[7] M. Wooldridge and N. R. Jennings, "Pitfalls of agent-oriented development," in *Proceedings of the Second International Conference on Autonomous Agents*, ser. AGENTS '98. New York, NY, USA: ACM, 1998, pp. 385–391. [Online]. Available: http://doi.acm.org/10.1145/280765.280867

[8] J. Himoff, P. Skobelev, and M. Wooldridge, "Magenta technology: Multi-agent systems for industrial logistics," in *Proceedings of the Fourth International Joint Conference on Autonomous Agents and Multiagent Systems*, ser. AAMAS '05. New York, NY, USA: ACM, 2005, pp. 60–66. [Online]. Available: http://doi.acm.org/10.1145/1082473.1082805

[9] J. Hendler, "Agents and the semantic web," *IEEE INTELLIGENT SYSTEMS*, vol. 16, no. 2, pp. 30–37, 2001.

[10] P. Flach, *Machine Learning: The Art and Science of Algorithms That Make Sense of Data*. New York, NY, USA: Cambridge University Press, 2012.

[11] R. Garreta and G. Moncecchi, *Learning Scikit-learn: Machine Learning in Python*. Packt Publishing, 2013.

[12] Y. Bengio, I. J. Goodfellow, and A. Courville, "Deep learning," 2015, book in preparation for MIT Press. [Online]. Available: http://www.iro.umontreal.ca/ bengioy/dlbook

[13] Y. LeCun and Y. Bengio, "word-level training of a handwritten word recognizer based on convolutional neural networks," in *Proc. of the International Conference on Pattern Recognition*, IAPR, Ed., vol. II. Jerusalem: IEEE, October 1994, pp. 88–92.

[14] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems*, p. 2012.

[15] R. M. Bell and Y. Koren, "Scalable collaborative filtering with jointly derived neighborhood interpolation weights," in *IEEE International Conference on Data Mining (ICDM*, 2007.

[16] L. A. Zadeh, "Fuzzy sets," *Information and Control*, vol. 8, pp. 338–353, 1975.

[17] E. Sanchez, "Resolution of composite fuzzy relation equation," *Information and Control*, vol. 30, pp. 38–48, 1976.

[18] M. Thomason, "Convergence of powers of a fuzzy matrix," *Journal of mathematical analysis and applications*, vol. 57, no. 1, pp. 476–480, 1977.

[19] H. J. Zimmermann, "Fuzzy programming and Linear Programming with several objective functions," *Fuzzy Sets and Systems*, vol. 1, no. 1, pp. 45–55, 1978.

[20] K. Atanassov, "Intuitionistic fuzzy sets," *Fuzzy Sets and Systems*, vol. 20, pp. 87–97, 1986.

[21] V. Kreinovich, A. Lakeyev, J. Rohn, and P. Kahl, *Computational Complexity and Feasibility of Data Processing and Interval Computations*. Kluwer Academic Publishers, 1998, ch. 10.

[22] J. M. Mendel, *Uncertain Rule-Based Fuzzy Logic Systems: Introduction and New Directions*, N. Upper Saddle River, Ed. Prentice Hall, 2001.

[23] D. H. Wolpert and W. G. Macready, "No free lunch theorems for optimization," *Trans. Evol. Comp*, vol. 1, no. 1, pp. 67–82, Apr. 1997. [Online]. Available: http://dx.doi.org/10.1109/4235.585893

[24] K. Pešková, J. Šmíd, M. Pilát, O. Kazík, and R. Neruda, "Hybrid multi-agent system for metalearning in data mining," in *Proc. of the MetaSel@ECAI 2014*, ser. CEUR Workshop Proceedings, J. Vanschoren, P. Brazdil, C. Soares, and L. Kotthoff, Eds., vol. 1201. CEUR-WS.org, 2014, pp. 53–54. [Online]. Available: http://ceur-ws.org/Vol-1201/paper-13.pdf

[25] A. E. Eiben and J. E. Smith, *Introduction to Evolutionary Computing*. SpringerVerlag, 2003.

[26] O. Kazík, J. Šmíd, and R. Neruda, "Evolutionary optimization of meta data metric for method recommendation," in *Cybernetics and Intelligent Systems (CIS), IEEE Conference on*, Nov 2013, pp. 123–127.

[27] R. Vilalta, C. G. Giraud-Carrier, P. Brazdil, and C. Soares, "Using meta-learning to support data mining," *IJCSA*, vol. 1, no. 1, pp. 31–45, 2004.

[28] J. R. Koza, *Genetic programming: on the programming of computers by means of natural selection*. Cambridge, MA, USA: MIT Press, 1992.

[29] T. Kren, M. Pilt, K. Peskov, and R. Neruda, "Generating workflow graphs using typed genetic programming." in *MetaSel@PKDD/ECML*, ser. CEUR Workshop Proceedings, J. Vanschoren, P. Brazdil, C. G. Giraud-Carrier, and L. Kotthoff, Eds., vol. 1455. CEUR-WS.org, 2015, pp. 108–109.